



Speech Therapy Suite

Therapist & Caregiver User Guide

A customisable speech-language practice suite for patients
and the clinicians and caregivers who support them.

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Introduction

The **Speech Therapy Suite** is a customisable, browser-based practice tool designed for people with acquired or progressive speech and language disorders. It supports a wide range of conditions — including aphasia, Primary Progressive Aphasia (PPA), post-stroke language difficulties, traumatic brain injury (TBI), and dementia-related communication decline — and can be tailored by a clinician or caregiver to match each patient's specific goals and needs.

The suite provides thirteen structured practice modules, an AI therapist assistant (Dr. Aria, powered by Claude), progress tracking, and export/import tools for clinician-customised content. Word lists, scripts, video clips, and practice tasks are all editable — making it straightforward to tailor sessions to any patient.

Modules at a Glance

Module	Purpose
AI Therapist (Dr. Aria)	Free conversation and encouragement with Claude AI
Assessment	Structured language evaluation tasks
Naming Practice	Picture/word naming with spaced repetition and cue ladder
Repetition	Repeat words and phrases at increasing difficulty
Script Training	Practise functional scripts for daily life
Sentence Builder	Drag-and-drop sentence construction
Sentence Work	Sentence completion and construction prompts
Video Questions	Comprehension questions on short video clips
Family & Friends	Graph-based family tree with personalised relationship labels
Memory	Multiple-choice memory questions drawn from family facts
Progress	Session history, accuracy trends, and SR statistics
Photo Library	Media URL library — images, videos, and audio with health checking
Profile	Patient name, daily goal, day streak, and condition type selector

Who is this guide for?

This guide is written for **speech-language pathologists (SLPs)** and **caregivers** who set up and supervise sessions. The interface is also designed to be used directly by the patient, with or without a helper present.

Sessions can be as short as 10–15 minutes. The app saves progress automatically in the browser — no account or internet connection is required for most features (an internet connection is only needed for the AI Therapist and for AI feedback within modules).

Installation

The app is distributed as a zip archive containing an installer script, the application bundle, and this documentation. Unzip the archive, then follow the steps for your operating system below. All files must remain in the same folder.

macOS

Requirements: macOS 12 Monterey or later · Node.js 18+ (the installer will guide you if missing) · Internet connection (for npm packages during install).

1. Double-click **Install PPA Therapy.command**.

If macOS shows a security warning: right-click the file → **Open** → **Open** to approve it once.

2. Follow the on-screen prompts to choose an install location and optionally enter your Anthropic API key. You can skip the key — the app will prompt for it on first launch.

3. When finished, double-click **Launch PPA Therapy** on your Desktop. The app opens automatically in your browser.

Windows

Requirements: Windows 10 or 11 · Node.js 18+ (the installer will guide you if missing) · Internet connection (for npm packages during install).

1. Double-click **Install PPA Therapy.bat**.

If Windows SmartScreen shows a warning: click **More info** → **Run anyway**. The script only creates a local project — it does not modify system settings.

2. Follow the on-screen prompts to choose an install location and optionally enter your Anthropic API key. You can skip the key — the app will prompt for it on first launch.

3. When finished, double-click **Launch PPA Therapy** on your Desktop. A console window opens briefly and your browser loads the app automatically.

Updating

To install a newer version, run the installer again with the updated bundle and choose the same install folder. Your session data and custom content are stored in the browser and are not affected. Download a backup from **Progress** → ■ **Settings** before updating as a precaution.

Uninstalling

macOS: Delete the install folder (default: `~/PPA Therapy`) and the Desktop launcher.

Windows: Use *Start* → *PPA Therapy* → *Uninstall PPA Therapy*, or delete `%LOCALAPPDATA%\PPA Therapy` and the Desktop shortcut manually.

Getting Started

Starting the App

Double-click **Launch PPA Therapy** on your Desktop. A small console window opens (this is normal — it runs the local server), and the app loads automatically in your browser within a few seconds. The recommended screen size is a tablet or laptop (10" or larger).

To stop the app, close the console window. The browser tab can be closed at any time — your progress is saved automatically.

iPad / direct browser use: The app can also be opened directly in Safari or Chrome without the installer — contact your clinician for the URL. On iPad, tap the Share button in Safari then **Add to Home Screen** to install it as a PWA and open it full-screen without the browser toolbar.

First-Time Setup (SLP / Admin)

The first time you open the app, everything is ready to use with default content. To customise the word lists and materials:

- Open the **Profile** module to set the patient's name and condition type (protected by the Caregiver PIN, default: 0000 — change this before clinical use).
- In the Naming module admin panel, add, edit, or remove picture cards.
- In other modules, tap the admin toolbar to edit word banks and prompts.
- Use **Export / Import** to save customised content as .ppa files that can be shared with other devices.

Navigation

The **sidebar** on the left lists all thirteen modules. Tap any module name to switch. Your place within a session is preserved when you switch and return.

API Key (AI features)

Dr. Aria and the AI feedback in each module require an **Anthropic API key**. Without a key the modules still work — they simply skip the AI feedback step and advance automatically. To add a key:

- Open Admin → Settings.
- Paste your Anthropic API key in the API Key field.
- The key is stored only in this browser — it is never sent to any server other than Anthropic.

Module Guides

AI Therapist (Dr. Aria)

Dr. Aria is a friendly, encouraging AI therapist powered by Claude. Use this module for open-ended conversation practice, warm-up chat, or emotional check-ins at the start of a session.

- Type a message in the box at the bottom and press **Send** (or Enter).
- Dr. Aria responds in natural, supportive language.
- There is no time limit — take as long as you need.

Assessment

Use this module for structured language evaluation. Tasks include single-word repetition, picture description, and sentence-level prompts.

- Select a task category from the list.
- Read the prompt aloud to the patient.
- Record the response using the text box or voice notes.
- Tap **Next** to advance.
- Tap the picture to enlarge it to twice its size. Tap outside the picture or press **Escape** to close.

Naming Practice

A picture or word card is shown. Try to name it. If the word is on the tip of your tongue, press the **Space bar** while the answer box is empty for a sound-starter hint:

- **Sound hints on (■)**: each press reveals and speaks the beginning of the word up to the next natural vowel sound, plus the following consonant — for example the first press on *glasses* shows and says **GLAS**; on *chair* it shows and says **CHAIR**. The pronunciation always matches how the word really sounds in context, not how the letters would sound in isolation.
- **Sound hints off**: each press reveals one letter group at a time with no audio.

The ■ button in the bottom-right corner of the hint area toggles sound hints. Your preference is saved automatically.

For more support, use the cue buttons:

- **Semantic Cue** — shows a written description or category clue (e.g. "a fruit").
- **Phonemic Cue** — shows the first sound of the word in square brackets (e.g. [g]).
- **Reveal** — shows the full word. Use this if no other cue has worked.

Answer box note: if there is already text in the box, the Space bar adds a space character rather than giving a hint — clear the box first if you want a hint.

When the patient types their answer, tap **Got it!** — or press **Enter / Return** — to record a correct response and get Dr. Aria's feedback. Pressing Enter also works in the Semantic Cue and Phonemic Cue input phases. Tap **Next word** → after Dr. Aria responds (or immediately if AI is not connected).

Keyboard navigation: press **Tab** to move between the practice buttons. The currently focused button is highlighted with a coloured ring. Pressing **Enter** while a button is focused activates it.

Tap the picture card graphic to enlarge it — useful if the image is small or the patient needs a closer look. Inside the zoom overlay: tap **+** or **-** to increase or decrease the size; drag the bottom-right corner handle to

resize freely. Tap outside the image or press **Escape** to dismiss the overlay.

■ Spaced Repetition — how words are scheduled

The Naming module uses a PPA-adapted spaced repetition engine. Words you recall easily are shown less often; words that need more support are shown more often. The maximum interval is 5 days — every word is reviewed at least weekly.

Result scoring:

correct (no cues used) $\times 1.2$ interval

space_cued (Space-bar hint used) $\times 0.9$ interval

semantic_cued $\times 0.8$ interval

phonemic_cued $\times 0.5$ interval

failed (full reveal) reset to 1-day interval

Words scored phonemic_cued or failed are re-queued 4 positions ahead for a second attempt within the same session.

■ Admin — customising the word list (Standard mode)

Add item: type a word; AI looks up an emoji and fills in cues.

Bulk import photos: select multiple images; AI fills empty word fields.

Generate by category: enter a category name and count; AI generates nouns.

Edit / Delete: change any field or remove cards. Export list as a .ppa file.

* Personal Photos mode — family and personal content

Tap the ■ My photos tab at the top of the Naming module to switch to a separate word list built from the patient's own photographs. This mode is caregiver-accessible — no PIN is needed.

Tap the pencil icon to open the Personal Library:

+ Add photo: choose an image, name the item, set a category and cues.

Bulk import: drag-and-drop or select multiple photos at once.

Edit / Delete: manage existing personal items.

Spaced repetition runs independently for personal items. Switching between Standard and My photos tabs preserves each mode's own SR progress.

Repetition

Structured repetition drills at five levels of difficulty — from single words through to full sentences. The module supports text-to-speech (TTS) playback and optional speech recognition so patients can practise aloud independently.

Basic use

- Choose a difficulty level using the tabs at the top (1 = single words, 5 = full sentences).
- The prompt is shown on screen.
- Tap **Play** to hear the item spoken aloud via text-to-speech.
- The patient repeats the word or phrase aloud.
- Tap **Correct** or **Incorrect** to record the result.
- Tap **Next** → to advance to the next item.

Speech recognition

Enable the microphone toggle in the settings row to let the patient speak their attempt and have it scored automatically:

- Tap **Start Speaking** — the button pulses while the microphone is active.
- A live transcript appears below the prompt as the patient speaks.
- The result is colour-coded: teal = great match ($\geq 75\%$ of words), amber = partial match ($\geq 40\%$), red = keep trying ($< 40\%$).
- The manual **Correct / Incorrect** buttons remain visible at all times and can always override the automatic score.

Voice and speed settings

A settings row above the level tabs provides a voice selector, a speed slider, and the speech-recognition toggle. Voice options depend on the browser — Chrome and Edge on desktop typically offer several natural-sounding English voices.

i Tip — enabling independent practice

Enable the microphone toggle and the patient can practise each item without a helper watching the screen. The automatic scoring records the attempt; the manual **Correct / Incorrect** buttons are always available to override it. Switching the level tab resets the microphone state for the new level.

Script Training

Functional scripts are short, high-frequency phrases for daily life — ordering food, greeting someone, answering the phone. Repeated practice builds automatic retrieval.

- Select a script from the list.
- The script is shown line by line.
- The patient reads or repeats each line.
- Tap **Next line** to advance.
- Dr. Aria can provide feedback at the end of a script if AI is connected.

Sentence Builder

A visual drag-and-drop tool for constructing sentences. Word tiles are arranged into categories (nouns, verbs, adjectives, etc.) and dragged into a sentence tray.

- Tap or drag a word tile from the word bank into the sentence tray.
- Rearrange tiles by dragging them within the tray.
- Tap **Check sentence** to get AI feedback.
- Tap **Clear** to start again.
- Filter the word bank by category using the tabs at the top.

Sentence Work

Two sub-modes for sentence-level language practice, selected by the tabs at the top. After each response, Dr. Aria provides written clinical feedback if AI is connected.

Sentence Completion

A sentence stem is displayed (e.g. *"Every morning I like to..."*) along with a topic hint. The patient types the rest of the sentence in the text box.

- Type a completion in the text box.
- Tap **Get Feedback** to receive Dr. Aria's commentary on grammar, meaning, and fluency.
- Tap **Skip** → to move on without feedback.
- Tap **Next task** → to advance after feedback.

Sentence Construction

A set of word chips is displayed alongside a hint. The patient builds a sentence by tapping chips or typing freely. **Not every word needs to be used** — each set is deliberately larger than the minimum sentence so that multiple valid answers are always possible.

- Tap a word chip to insert it at the cursor position in the text box.
- Tapping the same chip again inserts it a second time.
- Tap **Delete word** to remove the last word from the text box.
- Type additional words directly at any time — you are not limited to the chips.
- Tap **Get Feedback** — Dr. Aria accepts any grammatically correct sentence that uses some or all of the given words.

* Adaptive word ordering

The order the word chips appear adjusts automatically to the patient's measured ability. Three display levels are used:

Supported (purple badge) — words appear in sentence order. Maximum scaffold.

Standard (amber badge) — first half of words lightly shuffled.

Challenge (teal badge) — all words randomised. No positional cues.

After Dr. Aria's feedback, rate the difficulty using the three buttons:

Hard — lowers the difficulty level for the next task.

OK — keeps the current level and records today's date.

Easy — raises the difficulty level for the next task.

The step sizes are tuned to the patient's diagnosis profile: PPA adapts more conservatively than acute aphasia. If several days pass without practice, the stored level automatically decays toward Supported so the patient is not over-challenged at the next session.

+ Bonus Challenge — making a second sentence

After Dr. Aria responds, an amber Bonus Challenge card appears. The patient can try to construct a second, different sentence from the same word chips.

Dr. Aria evaluates the bonus sentence separately — confirming whether it is valid and genuinely different from the first. Tap Skip to go straight to the difficulty rating buttons.

The Hard / OK / Easy rating buttons only appear once the bonus is either attempted or skipped, so the difficulty signal reflects the full practice round.

i Admin — adding and editing tasks

Sentence Completion: type the sentence stem (e.g. 'When I feel happy, I...') and an optional topic hint.

Sentence Construction: type a complete example sentence in the sentence field (e.g. 'The dog runs in the park'). Words are extracted automatically — there is no need to comma-separate them. Add synonyms or alternate words to the same field to give the patient more flexibility and enable multiple valid answers (e.g. 'The dog runs in the park a big').

Export / Import: save custom tasks as .ppa files to share across devices.

Video Questions

Short video clips are played, followed by comprehension questions. This module supports auditory and visual processing alongside language. Two modes are available via the tabs at the top:

Standard mode (■) — built-in clips from YouTube plus any custom clips added via the admin panel. Managed with the caregiver PIN.

Personal mode (■■■) — clips uploaded from the device (local video files). Accessible with the caregiver PIN via the pencil icon. Video data is stored in the browser's IndexedDB, not localStorage, so it survives browser cache clears.

Watching a clip:

- Select a video from the list.
- Watch the clip (plays embedded in the app).
- Answer comprehension questions — multiple choice or open-ended.
- Tap **Submit** to record the answer.
- Tap **Next question** to continue.

Family & Friends

Build a family tree that personalises memory and naming exercises. The module stores people and their direct relationships — parent, partner, or sibling — and automatically derives all other labels (grandparent, aunt/uncle, cousin, etc.) so you never need to enter them manually.

- Tap **+** **Add** to add a person. Enter their name and optionally a photo URL.
- Under **Relationships**, choose the direction (parent of / child of / partner of / sibling of) and select the other person. Add as many links as needed.
- Under **Facts**, add items like birthday, job, or favourite colour — these are used by the Memory module.
- Tap **■ Tree** to switch to the family tree view. The patient node (You) always appears at the centre.
- Teal lines show parent–child connections; rose double-lines show partners; dashed lines show siblings when no shared parent is recorded.

Sticky default: when adding a new person, the "related to" selector starts on You (the patient). Once you change it to another person, that person becomes the new default for subsequent additions — useful when adding several members of one branch of the family.

Memory

The Memory module generates multiple-choice questions from the facts you have entered for family members and friends. For example, if Gran's birthday is recorded as "12 March 1942", the module might ask "When is Gran's birthday?" with four realistic options.

- Up to 8 questions per session, cycling through all family members with facts.
 - Wrong-answer distractors are drawn from other family members' facts, making the task meaningfully challenging.
 - Questions become harder (fewer obvious distractors) as accuracy improves.
 - Add more facts in Family & Friends to increase the variety of questions.
-

Progress

The Progress module shows a summary of all recorded activity.

Session log — a chronological list of activities from the current session.

Accuracy trends — correct/incorrect rates by module over time.

Naming SR statistics — per-word spaced repetition data: interval, streak, last result, next due date.

Progress data is stored locally in the browser. Use **Export Backup** to save a .ppabak file as an off-device copy.

Admin Panel

Content management areas in the app are protected by a 4-digit **Caregiver PIN** (default: **0000**). Change this in the **Profile** module before clinical use. The PIN prevents accidental edits by the patient and can be updated at any time without editing any source files.

Naming Module Admin

- **Add item** — type a word; the app will look up an emoji automatically (internet required), or you can upload an image.
- **Bulk import photos** — drag-and-drop or select multiple images at once. A grid shows each photo with fields for word, category, semantic cue, and phonemic cue. Blurring the Word field auto-fills the other three via AI. "Generate AI defaults for all" processes every row at once. Only rows with all fields filled are saved; words already in the list are skipped.
- **Generate by category** — type a category name and choose a count (default 15, max 50). The app calls the AI to generate the most common nouns in that category with emoji, semantic cue, and phonemic cue pre-filled. New items are pre-selected; duplicates are shown with their existing values and skipped automatically.
- **Edit item** — change the word, graphic, semantic cue, or phonemic cue.
- **Delete item** — removes the card from the practice list.
- **Export list** — saves the current list as a .ppa file.
- **Import list** — loads a .ppa file, merging with or replacing the current list.

Other Module Admin

Each module with editable content shows an admin toolbar at the top of the screen when Admin mode is active. Use it to add, edit, and delete items.

! Caregiver PIN security

The default Caregiver PIN is 0000. Change it in the Profile module before clinical use. The PIN prevents accidental content edits by the patient — it is not a strong security feature. No source code changes are needed: the PIN is stored in the browser and can be updated at any time.

Backup & Restore

All app data is stored in the browser's **localStorage**. This means:

- Data persists between sessions on the same device and browser.
- Clearing browser data or switching browsers will lose data.
- Data does not sync across devices automatically.

To back up:

- Open Admin → Backup & Restore.
- Tap **Export Full Backup** to download a .ppabak file.
- Store the file in a safe location (e.g. a shared folder or email it to yourself).

To restore:

- Open Admin → Backup & Restore.
- Tap **Import Backup** and select the .ppabak file.

- The page will reload with all data restored.

■ **Tip — backup after every customisation session**

Whenever you add new picture cards, edit scripts, or adjust word banks, export a fresh .ppabak backup. This protects your work if the browser data is ever cleared.

Profile

The Profile module stores information about the patient and controls how the app adapts over time. It is accessible to everyone — no PIN is needed to view it — but the condition type selector is protected by the Caregiver PIN.

Patient information

- **Name** — the patient's preferred name, shown as a greeting at the top of the Profile screen.
- **Daily goal (minutes)** — the target practice time per day (default: 20 minutes).
- **Day streak** — the number of consecutive days with at least one practice session. Resets if a day is missed.
- Tap **Save Profile** to apply changes to name and daily goal.

Condition type (Caregiver PIN required)

Tap **Change (Admin)** and enter the Caregiver PIN to reveal the condition selector. Choose the profile that best matches the patient's diagnosis:

Condition	Difficulty decay
Primary Progressive Aphasia (PPA)	7 days — fast decay; conservative re-entry
Chronic Aphasia	14 days — moderate decay
Acquired/TBI Aphasia	21 days — slower decay
Acute Aphasia	30 days — minimal decay
Dementia	5 days — very fast decay; maximum scaffold

The condition profile controls how quickly the adaptive difficulty level in modules such as Sentence Work decays back toward the Supported level when the patient has not practised for several days. A shorter half-life means the difficulty resets sooner, protecting the patient from being over-challenged at the next session. Select the profile that best matches the patient's diagnosis — it can be changed at any time as the patient's condition evolves.

Photo Library

The Photo Library is a personal media URL collection — a place to store links to images, videos, and audio clips that are meaningful to the patient. Items are labelled, tagged, and can be checked for URL health automatically. No PIN is required; the library is open to the patient and caregiver alike.

Adding items

- Tap **+ Add item** and enter a label and a URL (image, video, or audio link).
- Choose a media type: **Image**, **Video**, or **Audio**.
- Add one or more tags to help organise the library: person, family, animal, pet, place, food, object, plant, vehicle, or other.
- Tap **Save** to add the item to the library.

URL health checking

Each item shows a status indicator next to its label:

Status	Meaning
Reachable (teal)	URL responded with a success code — media is accessible
Not found (red)	URL returned a 404 or similar error — link may be broken
Unverified (amber)	URL was checked but the response was ambiguous
Unchecked (grey)	URL has not been checked yet in this session

Tap **Check all URLs** to run a health check on every item in the library. Individual items can be re-checked by tapping the status dot.

All items are stored in the browser (fam_content_items localStorage key) and are included in .ppabak backups.

Tips for Effective Sessions

Session length and pacing

Shorter, more frequent sessions are generally more effective than long infrequent ones. Aim for 15–30 minutes per day if possible, adjusted to the patient's stamina and engagement. Watch for signs of fatigue — increased errors, withdrawal, or frustration — and end the session early if needed.

Choosing the right modules

Begin sessions with a warm-up activity (AI Therapist or a well-known Script) before moving to more demanding tasks (Naming, Sentence Work). End on a positive note with something the patient finds rewarding.

Using cues effectively

In Naming Practice, try to allow a generous response time (10–15 seconds) before offering a cue. Use the least-support cue first — Space-bar phoneme hint, then Semantic Cue, then Phonemic Cue, then Reveal. This hierarchy is reflected in the spaced repetition scoring.

Involving the patient in goal-setting

Where possible, involve the patient in deciding which words to practise and which scripts matter most to them. Personally meaningful content produces better motivation and retention.

Recording baseline data

Before starting intensive practice on a new word set, record a baseline probe in the Assessment module. Repeat the probe every 2–4 weeks to track maintenance and generalisation.

Caregiver involvement

Caregivers can support practice by:

- Sitting alongside and providing gentle encouragement (not corrections).
- Reading prompts aloud if the patient finds on-screen text difficult.
- Noting which words or scripts the patient found particularly difficult or easy.
- Exporting a session log to share with the patient's SLP.

Troubleshooting

Below are the most common issues and how to resolve them.

Problem	Solution
Dr. Aria does not respond	Check your API key in Admin → Settings. Make sure you have an internet connection. The module will still advance without a response.
Picture cards show ■ instead of an image	The emoji look-up requires an internet connection. Try again online, or upload a photo from the camera roll in Admin mode.
Space bar does not give a hint	Make sure the answer box is empty. If there is any text in the box the Space bar adds a space instead. Clear the box first.
All my word lists have disappeared	Browser localStorage may have been cleared. Restore from a .ppabak backup file in Admin → Backup & Restore.
The app is very slow	Close other browser tabs. On iPad, ensure Low Power Mode is off. The app runs best in Chrome or Safari on a modern device.
Export / Import does not work	File downloads may be blocked by a browser setting or pop-up blocker. Check browser permissions and try again.
Audio hints do not play (sound icon)	Check that the device is not muted and the browser has permission to play audio. On iOS, the silent switch must be off. Tap the ■ button to toggle audio hints.
Speech recognition does not work	The browser must have microphone permission. In Chrome/Edge: click the lock icon in the address bar and enable Microphone. On iOS/Safari, go to Settings → Safari → Microphone. Speech recognition requires an internet connection on some browsers.